National College of Ireland

Masters in Science in Data Analytics (MSCDAD_A, MSCDAD_B, MSCDAD_C)

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Analytics Programming and Data Visualization (H9APDV)

Assessment Type - Project (70%)

Administrative Data:

Weight: The assignment will be marked out of 100 and is worth **70%** of the final module grade.

Instructions:

- This is a **group assessment**, with each team comprising 2-3 members. All submitted work must represent the team's original effort. Plagiarism, collusion, or the uncredited use of AI-generated content is strictly prohibited.
- Deliverables:
  - **Final Project Report**: approximately **5,000 words** in IEEE conference format (PDF) (including all figures, tables and references). Use clear academic writing and include properly formatted in-text citations and references.
  - **Video Presentation**: an **MP4** video of up to **10 minutes** demonstrating the project workflow, data pipeline, visualizations, findings, and limitations. Include team names and student numbers on the opening slide/frame.
  - **Code Artefact**: a **ZIP/GZ** archive containing all source code, notebooks, configuration files, and minimal sample data needed to run the project. Include clear setup and run instructions.
  - **Individual Work Breakdown Report**: a short ($\leq$ 1 page) PDF per student outlining individual contributions.
- All references must follow the **IEEE** referencing style.
- All reports must be submitted in **PDF** format.
- Teams may be required to attend a **viva** to discuss their submission.

Turnitin: All report submissions will be electronically screened for evidence of academic misconduct (i.e., plagiarism and collusion).

AI Usage Policy:

- Students **must not** use AI tools to write or generate report content, figures, or code.
- AI tools may be used **only for brainstorming or idea generation**, and such use must be **clearly declared** in the cover sheet, describing exactly how the tool was used.
- Failure to disclose AI usage will be considered a **breach of academic integrity**.

Late Submissions: Late submissions will not be penalized if the student applied for an extension through NCI360 and it was approved.

Following learning outcomes are addressed:

#	Learning Outcome Description
LO1	Analyse, compare, contrast and critically evaluate the characteristics of programming languages, programming environments and database systems commonly utilised for data analytics solution implementation.
LO2	Critically assess the challenges associated with processing big data datasets and compare and contrast programming for big data vis-à-vis programming for conventional datasets.
LO3	Evaluate tools and techniques for managing the data pipeline and preparing data for further analysis through data wrangling, cleaning, and validation.
LO4	Evaluate, design and implement solutions for processing datasets by using key programming patterns and constructs for data analytics, relevant programming languages, and suitable database systems
L05	Critically evaluate and apply data visualization design principles and interaction strategies in order to effectively communicate information to both technical and non-technical audiences.

1 Project Overview

This project assesses your ability to design and implement a complete data analytics and visualization solution using appropriate programming languages, environments, and database systems. You will demonstrate your ability to handle complex datasets, apply analytical techniques, and communicate insights effectively through clear and meaningful visualizations.

2 Project Requirements

Your project must include the following key components. Each stage should be clearly implemented and discussed in your report. The report structure is provided in section 3.1.

1. Each team member must contribute at least one dataset.
   - At least one dataset must be **semi-structured** (e.g., JSON, XML, or API/web-scraped data).
   - Other datasets may be structured (CSV, SQL tables) or unstructured (text, logs, etc.).
   - Each dataset should contain at least **2,000 records** to allow meaningful analysis.
2. All datasets must be **programmatically stored** in suitable databases before processing.
   - Acceptable databases include:
     - **Relational (SQL):** *PostgreSQL, MySQL, SQLite, Microsoft SQL Server*
     - **NoSQL / Semi-structured:** *MongoDB, Cassandra or DataStax Astra DB*

   The workflow should include actual database interactions (insert, query, retrieve) rather than direct file-based analysis.
3. **Data Pre-processing and Transformation:** Perform programmatic cleaning, wrangling, and transformation of your data. Examples include handling missing values, merging datasets, cleaning columns, filtering, or feature extraction. All processing steps must be reproducible through code.
4. **Data Integration and Processed Storage:** After transformation, store the processed outputs again in a database appropriate for analysis. For example, raw JSON data can be stored in MongoDB and cleaned and structured tables can be stored in PostgreSQL. Using multiple database types (e.g., NoSQL and SQL) is encouraged to demonstrate understanding of different storage paradigms.
5. **Programmatic Analysis and Visualization:** Perform meaningful analysis to discover insights and visualise results effectively. Analytical work may include exploratory data analysis (EDA), correlation studies, time-series exploration, or basic predictive modeling. You may combine results into an **interactive dashboard** using Streamlit, Dash, or a similar framework.

   All charts must also appear separately in the report with clear captions, interpretations, and discussion.
6. **[Optional Extension] Data Pipeline Orchestration:** As an optional enhancement, students are encouraged to demonstrate how their entire workflow - from data ingestion to cleaning, transformation, storage, and visualization, can be automated as a reproducible data pipeline.

   You may use a workflow orchestration tool such as Apache Airflow, Dagster, or Prefect to define and execute your ETL stages programmatically. This component is not mandatory but will be considered strong evidence of advanced understanding of ETL pipelines and production data workflows.

Example Workflow: You might use Python to retrieve a semi-structured JSON dataset via an API, store it in MongoDB, perform transformations and aggregations, then export the processed data to PostgreSQL. You could then analyse the data in Python, create visualizations showing key trends or correlations, and present them through an interactive dashboard built with Streamlit.

Suggested Data Sources:

- `https://catalog.data.gov/dataset?res_format=XML`
- `http://aiweb.cs.washington.edu/research/projects/xmltk/xmldata/`
- `https://data.gov.ie/dataset?res_format=JSON`
- `https://catalog.data.gov/dataset?res_format=JSON`
- `https://data.worldbank.org/`

Please note that *Kaggle* datasets will not be accepted. Additional dataset suggestions will be made available on Moodle.

3 Deliverables

The objectives, methodology and results of your analysis should be presented in the form of a project report. This report should discuss the programming and data processing challenges that you encountered and the means and mechanisms you implemented to overcome these challenges.

The report should be around **5,000 words** in length (including references, figures, tables, etc.), should use appropriate academic style and referencing, and be presented in the IEEE conference format. Templates for Microsoft Word and LaTeXcan be downloaded from the IEEE[1].

3.1 Project Report

- **Abstract:** This should provide a summary of the project objectives, methods and results. Take a look at abstracts from papers in your literature review to get an idea of what constitutes a good or bad abstract.
- **Introduction:** Briefly explain the purpose, relevance and motivation of your project. State the research question(s) & objectives or problem you aimed to solve. Summarize your approach and what readers can expect from the report.
- **Related Work:** Discuss similar work or academic references that influenced your choices. Explain how your approach builds on or differs from these works. Note that this should be a critical evaluation. It should be more than a mere summary of the works and should discuss their limitations and implications
- **Data Processing Methodology:** This section should contain :
  - A detailed description of the underlying dataset(s) and your justification for choosing them.
  - Full descriptions and justifications of the data processing activities carried out, such as use of APIs, databases, etc.
  - Complete descriptions and justifications of the implemented data processing algorithms.
  - Justifications for the choice of technologies used, such as programming languages, libraries and databases.
  - Diagrams providing a visual overview of the data gathering, processing and analysis flow.
- **Data Visualization Methodology:** Describe the visualizations and analyses you created. Explain why each chart or dashboard element was chosen (e.g., to show trends, correlations, comparisons). Refer to relevant design principles (e.g., clarity, colour use, interactivity). Include key figures in the report with clear captions and discussion.
- **Results and Evaluation:** Here you should present the main findings and insights of your work, making appropriate use of figures, tables, etc. You should provide clear evidence of how the project objectives were met, ensuring that you discuss your research findings, their interpretations and implications. You should also discuss any challenges or unexpected patterns.

[1] `https://www.ieee.org/conferences/publishing/templates`

- **Conclusion & Future Work:** In this section you should detail what others can/could learn from your work. You should discuss your findings in the context of the research question(s) you elicited earlier. You should present the limitations of your work, i.e. this should be a critical self-evaluation. Lastly, you should suggest potential directions for future work. Typically you would describe what you would do differently or how you would extend your work if you had more time.
- **References:** List all sources using IEEE citation style. Include academic papers, datasets, APIs, and documentation used.

3.2 Project Presentation

You should create a video presentation (maximum 10 minutes) that summarises your project and serves as a discussion point for your work. The presentation should clearly explain what you did, how you did it, why you chose your approach, and what you discovered.

Use a short slide deck to support your explanation, and include the link to the video recording (e.g., YouTube, OneDrive, or Google Drive) on the first slide. Ensure that all lecturers have permission to view the video, and that all team members speak briefly to present their respective parts of the project.

3.3 Code Artefacts

You should create a `.zip` or `.gz` archive containing all assets such as program code, data and system configuration details.

- **Source Code:** All Python scripts and notebooks, organised into logical folders (e.g., `src/`, `data/`, `visualizations/`).
- **Configuration Files:** Database connection scripts, `.sql` setup files, or environment variables (`.env`, `.yaml`).
- Include a small subset or provide download instructions if the full dataset is large.
- **Dependencies:** A `requirements.txt` or `environment.yml` file listing all libraries and versions.
- **README.md:** with:
  - A short description of the project
  - Clear setup and execution instructions (e.g how to run scripts, connect to databases, or launch the dashboard)
  - Any dependencies or assumptions

Optional: Include any DAG or pipeline definition files if you implemented an orchestration tool (e.g., Airflow or Dagster).

All code should be well-documented, use consistent naming, and follow standard formatting (PEP8 for python (`https://peps.python.org/pep-0008/`)

3.4 Work Breakdown Report

Each student must submit an individual Work Breakdown Report (1 page maximum) describing their specific contributions to the group project. This report should:

- Identify the tasks or components you were primarily responsible for (e.g., data collection, database design, cleaning, analysis, visualization, etc.).
- Briefly explain how you carried out your work and any challenges you faced.
- Mention any collaborative areas where multiple team members contributed together.

The report should be written in clear, concise language and saved as a PDF named using your student number (e.g., *x12345678.pdf*). It must be submitted separately to the *Work Breakdown Report* submission link on Moodle.

4 Submission Requirements

The project carries 70% of the total module marks. Your submission must include the following:

- **Project Report** – A PDF file named `TeamX.pdf` with the project cover sheet and AI declaration duly filled, uploaded to the *Project Report* Turnitin link on Moodle. Include the names and student numbers of all team members on the front page.
- **Code Artefact** – A `.zip` or `.gz` archive named `TeamX.zip` (or `.gz`), uploaded to the *Code Artefact* link on Moodle.
- **Video Presentation** – Submit a short **slide deck (PDF or PPT)** containing the **link to your recorded video presentation** (e.g., YouTube, OneDrive, or Google Drive). The link must be placed on the **first slide**, and viewing permissions must allow all lecturers to access the video. The presentation should be no longer than **10 minutes**, and all team members must speak briefly to present their part.
- **Work Breakdown Report** – **Each student** must submit an individual report (maximum one page) outlining their contribution. The file should be named using your student number (e.g., `x12345678.pdf`) and uploaded to the *Work Breakdown Report* link on Moodle.

Late submissions will not be accepted unless an official extension is approved through NCI360.
Each team should elect one member to submit the report, code artefact, and presentation on behalf of the group.

5 Marking

The project will be marked according to the grading rubric provided in the last two pages of this document.

6 Academic Integrity

Any written work created by others must be properly cited and paraphrased or summarised where possible; otherwise, it should appear in quotation marks. Figures not created by you must include an acknowledgement specifying the name(s) of the creator(s). Code obtained from online sources must not be claimed as your own. Instead, include a comment in the source code clearly indicating the original source and author.

Use of AI: While AI tools may be used *only to locate related literature or data sources*, any other use of AI, including but not limited to generating text, code, figures, or visualizations is **strictly prohibited**. Any unacknowledged or undisclosed use of AI will be considered a breach of academic integrity.

Students are strongly advised to familiarise themselves with the Guide to Academic Integrity provided by the NCI Library.

Note: All submissions will be electronically screened for evidence of academic misconduct, including plagiarism, collusion, or misrepresentation. Any work found to contain such evidence will be referred to the College's Academic Misconduct Committee for disciplinary action.

Table 1: Grading Rubric - Analytics Programming & Data Visualization Project (Semester 1, 2025/26)

Criterion	**Solid H1 ($\geq$ 80%)**	**H1 (70% - 79%)**	**H2.1 (60% - 69%)**	**H2.2 (50% - 59%)**	**Pass (40% - 50%)**	**Fail ($<$ 40%)**
Project Objectives (10%)	Very challenging and well-defined project objectives are exceptionally clear, fully met, and thoroughly discussed. The project fully aligns with the brief, including $\geq$ 1 dataset per team member and $\geq$ 1 semi-structured dataset.	Challenging and well-presented objectives are fully met and discussed; brief requirements clearly met.	Reasonable objectives are well presented, fully met, and adequately discussed; partial adherence to brief evident.	Objectives clear and mostly met; some brief requirements (e.g., dataset criteria) partly addressed.	Unambitious objectives at least partially met and briefly discussed.	The objectives of the project are unclear, have not been discussed. It is not possible to discern if the objectives have been met.
Literature Review (10%)	An excellent critical analysis of substantive and highly relevant literature	A very good critical analysis of substantive and relevant literature.	A good analysis of relevant literature. The critical analysis aspect could be somewhat stronger.	An adequate analysis of mostly relevant literature. The critical analysis aspect could be significantly stronger	A limited analysis of some relevant literature but it lacks evidence of understanding	Little/no relevant literature; minimal understanding.
Data Complexity and Handling (15%)	Datasets exceptionally well prepared, explored, and stored before/after processing in appropriate databases. **Both datasets high complexity**; $\geq$ **1 semi-structured or programmatically retrieved** (API/JSON/XML/scraped). Thoughtful schema/collection design; strong integration.	Well prepared/explored; stored in suitable databases. **At least one** dataset complex and semi-structured.	Well prepared; stored in an appropriate database. **At least one** dataset non-trivial or moderately complex.	Adequately prepared; database usage evident but limited.	Basic/trivial datasets; storage approach may be inappropriate.	Only trivial datasets; no meaningful database storage/prep.

Criterion	Solid H1 ($\geq$ 80%)	H1 (70% - 79%)	H2.1 (60% - 69%)	H2.2 (50% - 59%)	Pass (40% - 50%)	Fail ($<$ 40%)
Data Processing (20%)	The data processing workflow is exceptionally well designed, efficient, and reproducible. All major ETL stages (extraction, transformation, loading) are implemented programmatically in Python using appropriate libraries. Clear evidence of automation, modular code, and effective handling of complex data issues (e.g., missing values, merging, reshaping). May also demonstrate optional pipeline orchestration (e.g., Airflow, Dagster).	Data processing steps are well implemented, logically structured, and central to the project objectives. Appropriate Python libraries are used effectively to clean, transform, and prepare data for analysis. Workflow is clearly reproducible and well documented.	Data processing is appropriate and functional, covering most ETL stages. Some steps could be better structured or documented. Code achieves results but may lack optimisation or full automation.	Meaningful but limited data processing. Some manual steps remain or code structure is unclear. Evidence of basic cleaning and transformation, but limited reproducibility.	Basic or partial implementation of data processing steps. Minimal cleaning or transformation performed. Heavy reliance on manual operations.	Little or no data processing implemented. Code is missing, incomplete, or demonstrates a poor understanding of ETL concepts.
Data visualization (15%)	visualization choices are highly appropriate, exceptionally well presented and are fully justified using relevant theory	visualization choices are very appropriate, well-presented and are well justified using relevant theory	visualization choices are appropriate, adequately presented and are accompanied by a basic justification that draws on mostly relevant theory	visualization choices are somewhat appropriate, adequately presented but lack a solid justification using relevant theory	visualization choices appear to be random and are not justified using appropriate theory	Visualisaiton (if any) are very poorly conceived, are illegible and are not discussed in the context of appropriate theory
Results and Conclusions (20%)	Three or more insightful findings are excellently presented and thoroughly discussed in the context of the domain using appropriate references to prior work.	Three or more interesting and non-arbitrary findings are presented and thoroughly discussed in the context of the domain using appropriate references to prior work.	Three or more relevant findings; some interpretation.	Two or more interesting non-arbitrary findings are presented and appropriately discussed	Two or more interesting non-arbitrary findings are presented but are poorly discussed	Little to no non-arbitrary results and/or findings are presented

Criterion	Solid H1 ($\geq$ 80%)	H1 (70% - 79%)	H2.1 (60% - 69%)	H2.2 (50% - 59%)	Pass (40% - 50%)	Fail ($<$ 40%)
Quality of Writing (10%)	Exceptionally well written, with no language errors. All figures are well conceived, readable and correctly captioned. The IEEE template is strictly adhered to. The report does not exceed the length limits. All references are appropriately and correctly used	Well written, with no significant language errors. All figures are well conceived, readable and appropriately captioned. The IEEE template is adhered to. The report does not exceed the length limits. References are appropriately and correctly used	Well written, but has a few significant language or style errors. Figures are well presented. The IEEE template and length limit are adhered to. References are complete and correctly used	Adequately written, but has a few significant language and/or style errors. Some figures are hard to read. The IEEE template and length limit are mostly adhered to. References are complete, and correctly used	Adequately written, with some significant language and/or style errors. Figures may be hard to read or presented in a sub-optimal manner. The IEEE template may not have been followed. References are mostly complete and correctly used	Poorly written and littered with typographical errors and/or poor use of English. The IEEE template was not used. Figures may be hard to read. References (if any) are largely incomplete.